

Growing-Control Covariance Spectrum

Stable Energy, Noncanonical Signed Interaction

Liang Wang

School of Mathematics and Statistics

Huazhong University of Science and Technology (HUST), Wuhan, China

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Abstract

TPC-337 found that centered variation dominates the response of four masked source components under five coordinate controls. We enlarge that orbit to nine controls while holding the source and the all-plus operator fixed. The centered fraction remains high in every one of six windows (87.72–89.73 percent), and the trace-normalized covariance spectrum changes by only 0.0264–0.0441 in L^1 . A selected signed interaction is not stable: twin-zero covariance is negative for the five-control ensemble and positive for the nine-control ensemble in all six rows. Thus the finite energy structure is more reproducible than any signed covariance law.

1 Question and nested ensembles

Write the parent-locked source as

$$\beta = \sum_{C \in \{T, B, P, Z\}} \beta_C, \quad \beta(t) = \Lambda(t+2) - b^{(2)}(t),$$

on the six intervals $I_{o,N} = \{o, \dots, o + N/2 - 1\}$ with $o \in \{42001, 44001\}$ and $N \in \{2048, 4096, 8192\}$. The labels denote twin prime, non-twin prime shift, prime-power shift, and zero-support.

The fixed operator is the all-plus deleted-diagonal shell matrix

$$A = \sum_{54 < p \leq 108} B_{p,54,1}, \quad H = 66.$$

The five-control ensemble is identity, the affine maps (3, 11), (5, 17), (7, 29), and reversal. The nine-control ensemble adds (9, 1), (11, 13), (13, 17), and (17, 19) modulo the source count. All multipliers are odd, so these are bijections on the powers-of-two source counts.

For an ensemble J , define

$$y_{C,j} = AP_j \beta_C, \quad \bar{y}_{C,J} = \frac{1}{|J|} \sum_{j \in J} y_{C,j}, \quad z_{C,j,J} = y_{C,j} - \bar{y}_{C,J}.$$

2 Finite identity and spectrum

The centered vectors sum to zero. Therefore, for either nested ensemble,

$$\frac{1}{|J|} \sum_{j \in J} \|y_{C,j}\|_2^2 = \|\bar{y}_{C,J}\|_2^2 + \frac{1}{|J|} \sum_{j \in J} \|z_{C,j,J}\|_2^2. \quad (1)$$

The class covariance Gram matrix is

$$K_J(C, D) = \frac{1}{|J|} \sum_{j \in J} \langle z_{C,j,J}, z_{D,j,J} \rangle.$$

For any $a \in \mathbb{R}^4$,

$$a^T K_J a = \frac{1}{|J|} \sum_{j \in J} \left\| \sum_C a_C z_{C,j,J} \right\|_2^2 \geq 0.$$

This proves positive semidefiniteness but says nothing about the sign of an individual off-diagonal entry. Since changing J changes the centering, there is no nested sign monotonicity to assume.

We compare the eigenvalues of K_J after division by its trace. This removes the overall response scale for a descriptive finite spectral comparison; it does not create a limiting object.

3 Audit and exact anchor

The producer computes all nine outputs once per class and derives both nested ledgers. An independent checker rebuilds the source by trial factorisation and accumulates the prime shell in reverse order. Mutation stress tests the control count, energy dominance, sign reversal, anchor, and firewall. A rational two-vector anchor verifies the mean/centered identity exactly: for orbit vectors $(1, 0)$ and $(0, 1)$, the average, coherent, and centered energies are 1, $1/2$, and $1/2$.

4 Results

Table 1: Nested control-orbit readout over six windows.

quantity	five controls	nine controls
full centered fraction	0.785032–0.855298	0.877180–0.897264
full coherent fraction	0.144702–0.214968	0.102736–0.122820
twin/background covariance	+ in 6/6	+ in 6/6
background/zero covariance	– in 6/6	– in 6/6
twin/zero covariance	– in 6/6	+ in 6/6

The normalized covariance-spectrum L^1 distance between the two ensembles is 0.0264396–0.0440592. The corresponding unnormalized covariance matrices change by 0.217894–0.237903 in relative Frobenius norm. These two facts can coexist: the spectral shape is fairly close after normalization, while a particular off-diagonal entry crosses zero.

The sign reversal is not a numerical borderline phenomenon. The smallest absolute twin–zero entry in each ensemble is separated from zero by the declared guard, and the independent replay reproduces both six-row censuses. It nevertheless remains a finite statement about two explicitly chosen ensembles.

5 Interpretation and firewall

The positive result is a nested validation of the robust part of TPC-337: position-sensitive covariance carries most of the output response even after four new affine controls are added. The obstruction is that signed class interaction is not canonical. A proof strategy that needs a fixed sign for the twin–zero term cannot be justified by the five-control panel.

The identities are `PROVED_EXACT_FINITE_DECLARED_MODEL`; the nested replay, energy census, and finite spectral distances are `NUMERICALLY_CERTIFIED_FINITE`. The sign reversal is `REFUTED_SCOPED` for ensemble invariance. There is no arithmetic advance, fixed-power credit, official evaluator pass, or twin-prime result:

`ARITHMETIC_ADVANCE=NO,      FIXED_POWER_CREDIT=0,      FULL_GATE_B=OPEN.`

The official Session evaluator files are absent; local Bridge-B is fail-closed.

6 Next question

The next route should avoid relying on signed covariance. We therefore test a sign-free mask-aware operator envelope and compare it with the observed gains.